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9-13.03.2026

International Digital Twins in Astrobiology Hackathon

HACK-4-SAGES

Welcome webinar 09.03.2026

Co-funded by
the European Union

ENHANCE
European Universities of Technology Alliance

ETH zürich

GDAŃSK UNIVERSITY
OF TECHNOLOGY

POLITECHNIKA WARSZAWSKA
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Introduction

AGENDA

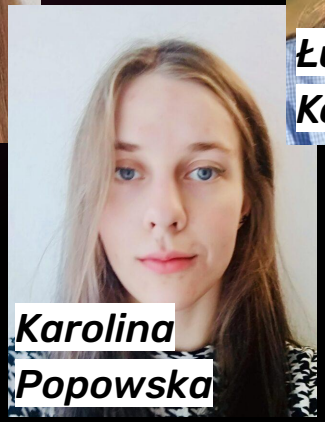
01 Introduction

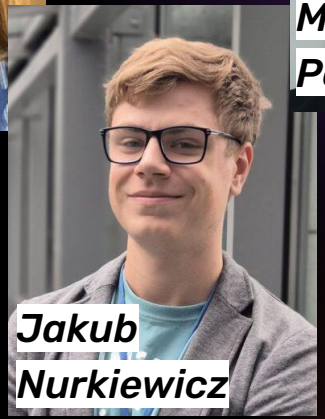
02 Schedule

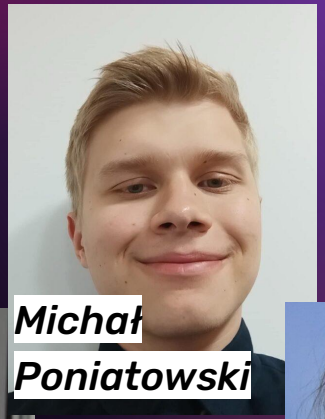
03 Q&A

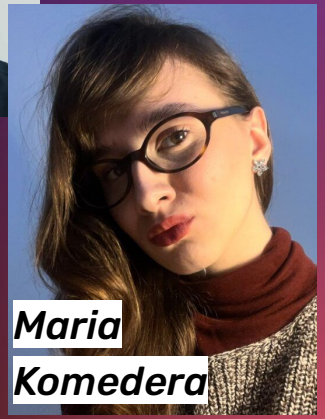
04 Welcome lecture by Dr. Antonin Affholder

Let me introduce us









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Congratulations!

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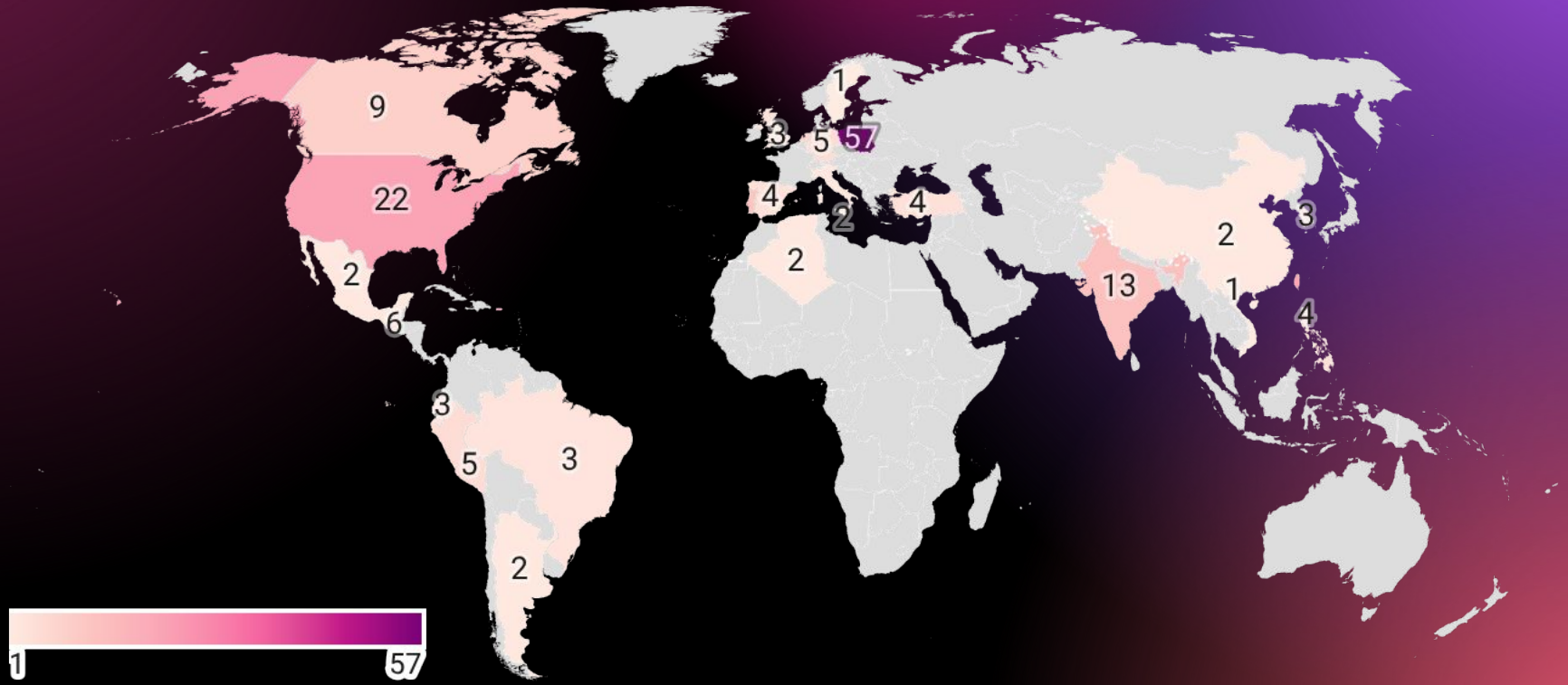
194 applicants across 5 continents!



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52 accepted teams

	Digital Twins in Origins of Life	Digital Twins in Life Detection Biosignatures	Digital Twins in Exoplanet Habitability
No. of accepted teams	13	19	20

Schedule

Schedule (UTC +1)

Time (Zurich)	9th March 2026	10th March 2026	11th March 2026	12th March 2026	13th March 2026
00:00	Mentor meeting	**Start of hackathon**			Jury deliberations
00:00 - 09:00					
09:00 - 10:00		Live Q&A	Live Q&A	Live Q&A	
10:00 - 12:00					
12:00 - 12:30		Live daily check	Live daily check	Live daily check	
12:30 - 17:00					
17:00 - 18:00	Welcome webinar				
18:00 - 19:00	Mentor meeting	Live Q&A	Live Q&A	Live Q&A	
19:00 - 20:00					Winners announcement
20:00 - 23:59					
00:00				**Submission deadline**	

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Mentorship

- Your mentors are available for 1 hour each day for you.
- You are responsible for communicating with your mentor. Please, create a Slack group with all your team members and your mentor to facilitate discussions.
- All mentors and teams received when2meet and google meets links via email, and shall schedule their meetings using these.
- Shout out to all our awesome mentors, thank you!

Q&As and daily checks

- Live Q&A sessions are optional. These are opportunities to talk with the organizers live and ask questions. They will not be recorded.
- Live daily check-ins will be short meetings where we summarize questions and concerns that came up during the day and provide clarifications. We recommend attending if possible, but participation is not mandatory. They will not be recorded.
- You are also always encouraged to ask questions via your designated Slack channels.

Project submission

- You must produce a max. 2 minute long video, and post it on youtube (ensure it's not set to private).
- Then you must submit it via our Google Form (sent via email and Slack) by 24:00 12th March UTC+1.
- The proper submission form includes university email addresses. If you don't have one, contact us asap.
- Only one person from the team shall submit the results.
- Winners will have to provide official proof of being BSc/undergraduate students.

Judging criteria

1. Fit to the Selected Competition Track
- Compliance with the selected category.
- Clarity of the scientific problem addressed within the chosen track
- Consistency between the project objectives and the thematic scope

2. Relevance to the Broader Astrobiology Field
- Scientific relevance beyond the specific use case
- Contribution to current challenges in astrobiology and planetary science
- Potential impact on future research, education, or space exploration activities

3. Innovation and Originality
- Level of innovation of the proposed solution
- Novelty of the concept, approach, or methodology
- Advancement beyond existing tools, models, or conceptual frameworks

4. Use of Digital Technologies and Digital Twins
- Effective use of digital technologies
- Appropriateness of the digital-twin concept for the problem addressed
- Technical coherence between scientific objectives and implementation

5. Academic Maturity and Problem-Solving Skills
- Clear formulation of the research question or problem
- Logical reasoning and structured methodology
- Awareness of assumptions, uncertainties, and limitations
- Evidence of interdisciplinary thinking and critical analysis

6. Reproducibility and Feasibility
- Transparency of methods, data sources, and assumptions
- Potential for the solution to be reproduced or extended by others
- Technical feasibility within realistic constraints
- Scalability or adaptability to other astrobiology applications

7. Quality of Project Presentation
- Clarity and functionality of the prototype or proof-of-concept
- Quality of visualizations, figures, or interfaces
- Structure and clarity of the presentation
- Effectiveness of communication to a multidisciplinary audience

Judges

Marie-Luise Steinmeyer, PhD

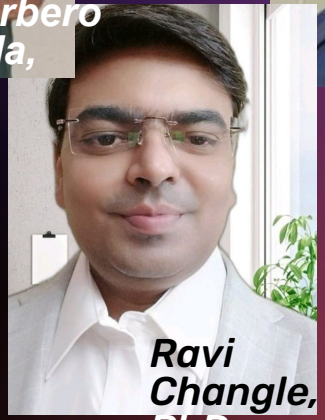
Dariusz Plewczynski, PhD

Borja Barbero Barcenilla, PhD

Carlo Tortora Brayda

Janusz Petkowiski, PhD

Jian Gong, PhD



Ravi Changle, PhD

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Awards

- Mentorship during the summer to give you a chance to improve your projects.
- Networking seminars with experts in the field.
- A paid trip to Switzerland to attend the Origins Federation Conference Early Career Day and present your winning projects there.
- Details will be communicated with the winners via email next week.

Any questions?

Q: What can be used and what cannot be used

You can use:

- Public datasets (Earth, planetary, climate, geochemical, biological)
- Published models or simplified versions of them
- Existing frameworks (e.g. climate, ocean, atmospheric, ecological models)
- ChatGPT and other GenAI

You cannot use:

- Proprietary or restricted data without permission
- Present results copied directly from a paper as “new”

Q: Could you please confirm if participants retain full 100% ownership of their code and ideas?

- **Yes!** These are your projects and ideas, and we do not claim any ownership over them.
- The only part of it that we want to have free access to are the videos you'll make, so if there are aspects of your projects you aren't comfortable sharing, they can't be mentioned in the videos.
- Videos of the winners and honorable mentions will be posted on our social media channels.

Q: My team had chosen to participate in person and therefore we would have in-person meetings with our mentor. However, from what I understood in one of the emails, our meetings will actually be held virtually, via Google Meet. Could you please confirm this for me?

- If you happen to be matched with a mentor from your in-person university, you are welcome to ask them to meet in person.
- You were matched with mentors based on their expertise, not location, though, so chances are high that your mentor is not from your university and therefore can't meet you in person.

Q: Can we meet with our mentor other days than monday?

- **Yes!** We suggest you to meet with your mentor for an hour each day (9th-12th March).
- Please, please, please read our emails and communicate your meeting times with your mentors via Slack.
- If you don’t want to meet with your mentor, inform them anyway, they’re expecting to hear from you!

Q: How to get connected to your mentor?

You received information via email about your assigned mentor and their availability. Follow these steps:

1. Select a 1 hour time-slot on each of the hackathon days via when2meet link. Make sure it doesn’t overlap with other teams.

2. Create a Slack group chat with your teammates and mentor.

3. Inform your mentor about your selection via this group chat.

4. Meet with your mentor on each of these days

Good luck!

Welcome lecture by Dr. Antonin Affholder

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Dr. Antonin Affholder

Image: Peter Pedersen / COPL

- Alma mater: Ecole Normale Supérieure Paris

- NOMIS–ETH Fellow at the Centre for Origin and Prevalence of Life at ETH Zurich (Switzerland)

- Astrobiologist working on habitability and biosignatures

- He developed models for the atypical microbial ecosystems, which may have dominated the biosphere of Earth’s young years and now populate the oxygenless deep ocean and soils